

Generative Market Simulation

Human Strategies & the News-Layer Design

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① Human Trading Strategies

② Environment Design

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Four Categories of Trading Strategies

- **Information-Driven:** Uncover the true probability via superior data and interpretation.
- **Behavior-Driven:** Exploit the systematic errors and predictable biases of other traders.
- **Structure-Driven:** Capitalize on price inconsistencies and market mechanics.
- **Game-Driven:** Influence or coordinate the market itself; play the game instead of predicting it.

* *Meta Strategies (change the market):* Create markets, create accounts — waiting for exploration.

Capitalize on informational asymmetry before market consensus.

Strategy	What it does
Fundamental pricing	Find the fundamental information, and trade when the market gets it wrong.
Informed trading	Acquire private signals; trade ahead of price discovery
News / event trading	React rapidly to novel information and its probability impact

Refs. Grossman–Stiglitz 1980 · Kyle 1985

Profit from predictable errors and behavioral biases of other traders, rather than fundamental truths.

Strategy

What it does

Trade "reverse stupidity"

Bet "No" on dramatic events, exploiting retail crowds who systematically overprice sensational or wishful outcomes.

Refs. Snowberg–Wolfers 2010 · Thaler–Ziemba 1988 · De Bondt–Thaler 1985

Extract profit from market mechanics and mathematical inconsistencies, without needing a directional view.

Strategy

What it does

**Cross-market
arbitrage**

Exploit temporary price gaps across logically correlated contracts (A , B , $A \vee B$)

Market making

Provide liquidity on both sides to earn the bid-ask spread and rewards

Refs. Law of One Price

Generate profit by actively coordinating behavior, distorting signals, or manipulating the market itself.

Strategy

What it does

Split attack

Deploy multiple accounts to fabricate an illusion of consensus or volume

Pump-and-dump

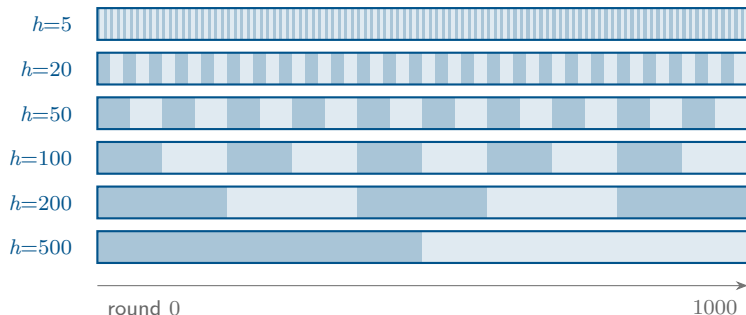
Artificially inflate prices to attract followers, then liquidate at the peak

Refs. Rhode–Strumpf 2007

① Human Trading Strategies

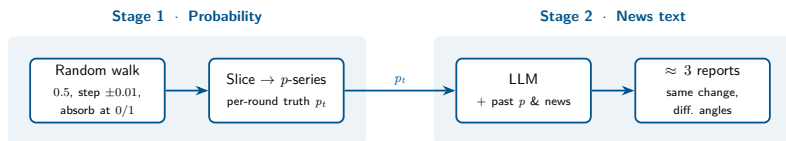
② Environment Design

- **Market setup** — 12 active markets across 6 timeframes (5/20/50/100/200/500), 1000 rounds
- **Auto-restart** — when a market ends, a new one of the same type opens; all timeframes are always available



shorter horizon $\Rightarrow$ more frequent restart; coverage never gaps

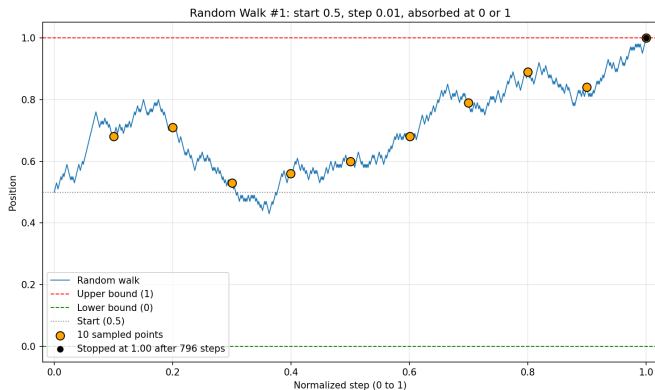
Two steps turn a hidden probability into readable news.



p is the input to the text, never written into it — the number itself stays in private signals.

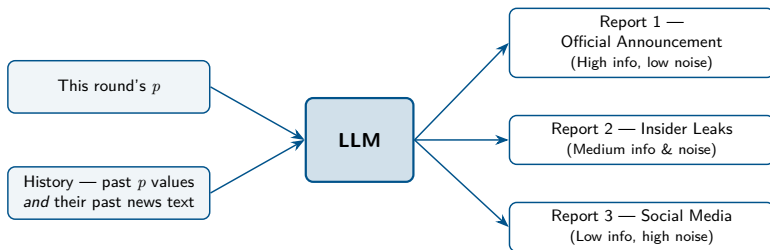
Stage 1 — where the probability comes from

A random walk produces the truth p ; we slice the finished path into per-round probabilities.



Refs. Samuelson 1965 · Wolfers–Zitzewitz 2004

Feed the probability to an LLM; it returns the round's news as a few faithful, different-angle reports.



Refs. Blackwell 1953

Thanks!

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